**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

MGU-UGP (HONOURS) REGULAR EXAMINATION – March 2026

**Fourth Semester**

**Skill Enhancement Course (SEC)**

**MG4SECCSC200 – FOUNDATIONS OF DATA SCIENCE**

**Duration: 1 Hour 30 Minutes** **Maximum Marks: 50**

**VALUATION SCHEME**

***Part A***

***Answer all questions***. *Each question carries* ***2 marks***

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| **1** | **Define Secondary Data.** | |
| Ans. | Data collected by others for a different purpose. | Definition - 1.5 Marks<br>Example - 0.5 Marks |
| **2** | **Find the Mode of: 4, 6, 6, 9, 11** | |
| Ans: | 6 | Correct Answer – 2 Marks |
| **3** | **Identify whether the events "getting head" and "getting tail" are mutually exclusive.** | |
| Ans: | Yes | Identification – 1 Mark<br>Reason – 1 Mark |
| **4** | **Describe the different types of averages used in statistics.** | |
| Ans: | Mean, Median, Mode | Name of Averages – 1 Mark<br>Definition of any 2 - 1 Mark |
| **5** | **Write the formula for conditional probability.** | |
| Ans: | $P(A\|B) = P(A \cap B)/P(B)$ | Correct Formula - 2 Marks |
| ***Part B***<br>*Short Answer Questions*<br>***Answer any 5 questions***. *Each question carries* ***5 marks*** | | |
| **6** | **Distinguish between diagrammatic presentation and graphic presentation of data.** | |
| Ans: | • Diagrammatic presentation uses diagrams like bar charts and pie charts.<br>• Graphic presentation uses graphs like line graphs and histograms. | Definition – 2 Marks<br>Differences (any 3 points) – 3 Marks |

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| **7** | **Write a brief note on central tendency and the purpose of average. Find the arithmetic mean of: 14, 18, 22, 26.** | |
| Ans: | Central tendency refers to the central value of a dataset (mean, median, mode).<br>Mean = (14+18+22+26)/4 = 80/4 = 20 | Explanation – 2 Marks<br>Formula – 1 Mark<br>Calculation – 2 Marks |
| **8** | **Define a random experiment. Explain the term clearly and give one suitable example.** | |
| Ans: | A random experiment is one whose outcome cannot be predicted with certainty.<br>Example: Tossing a coin. | Definition – 2 Marks<br>Explanation – 1 Mark<br>Example – 2 Marks |
| **9** | **Describe the direct personal investigation method of collecting primary data.** | |
| Ans: | In this method, the investigator collects data personally from respondents. | Definition – 2 Marks<br>Explanation – 2 Marks<br>Advantage/Disadvantage – 1 Mark |
| **10** | **Briefly explain Arithmetic Mean and find the average of the marks obtained by 6 students: 40, 45, 50, 55, 60, 65.** | |
| Ans: | Arithmetic Mean is the sum of observations divided by number of observations.<br>Mean = 315/6 = 52.5 | Definition – 1 Mark<br>Formula – 1 Mark<br>Calculation – 3 Marks |
| **11** | **State the addition theorem of probability for three events. Write the formula and explain the meaning of each term.** | |
| Ans: | $P(A \cup B \cup C) = P(A)+P(B)+P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$ | Formula – 3 Marks<br>Explanation – 2 Marks |
| **12** | **State the axioms of probability. Explain each axiom briefly.** | |
| Ans: | 1. $P(A) \geq 0$<br>2. $P(S) = 1$<br>3. Additivity for mutually exclusive events | Listing – 3 Marks<br>Explanation – 2 Marks |
| ***Part C***<br>*Essay Questions*<br>***Answer any 1 question***. *Each question carries* ***15 marks*** | | |
| **13** | **Distinguish between diagrammatic presentation and tabular presentation of data.** | |

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| Ans: | Diagrammatic presentation uses visual diagrams like bar charts and pie charts.<br>Tabular presentation uses rows and columns. | Introduction – 2 Marks<br>Diagrammatic – 4 Marks<br>Tabular – 4 Marks<br>Differences – 3 Marks<br>Conclusion – 2 Marks |
| **14** | **Define Mode. Explain its merits and demerits and compare it with Mean and Median.** | |
| Ans: | Mode is the value that occurs most frequently. | Definition – 2 Marks<br>Merits – 4 Marks<br>Demerits – 4 Marks<br>Comparison – 3 Marks<br>Conclusion – 2 Marks |